



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PARAMATHY-VELUR TALUK LOA

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

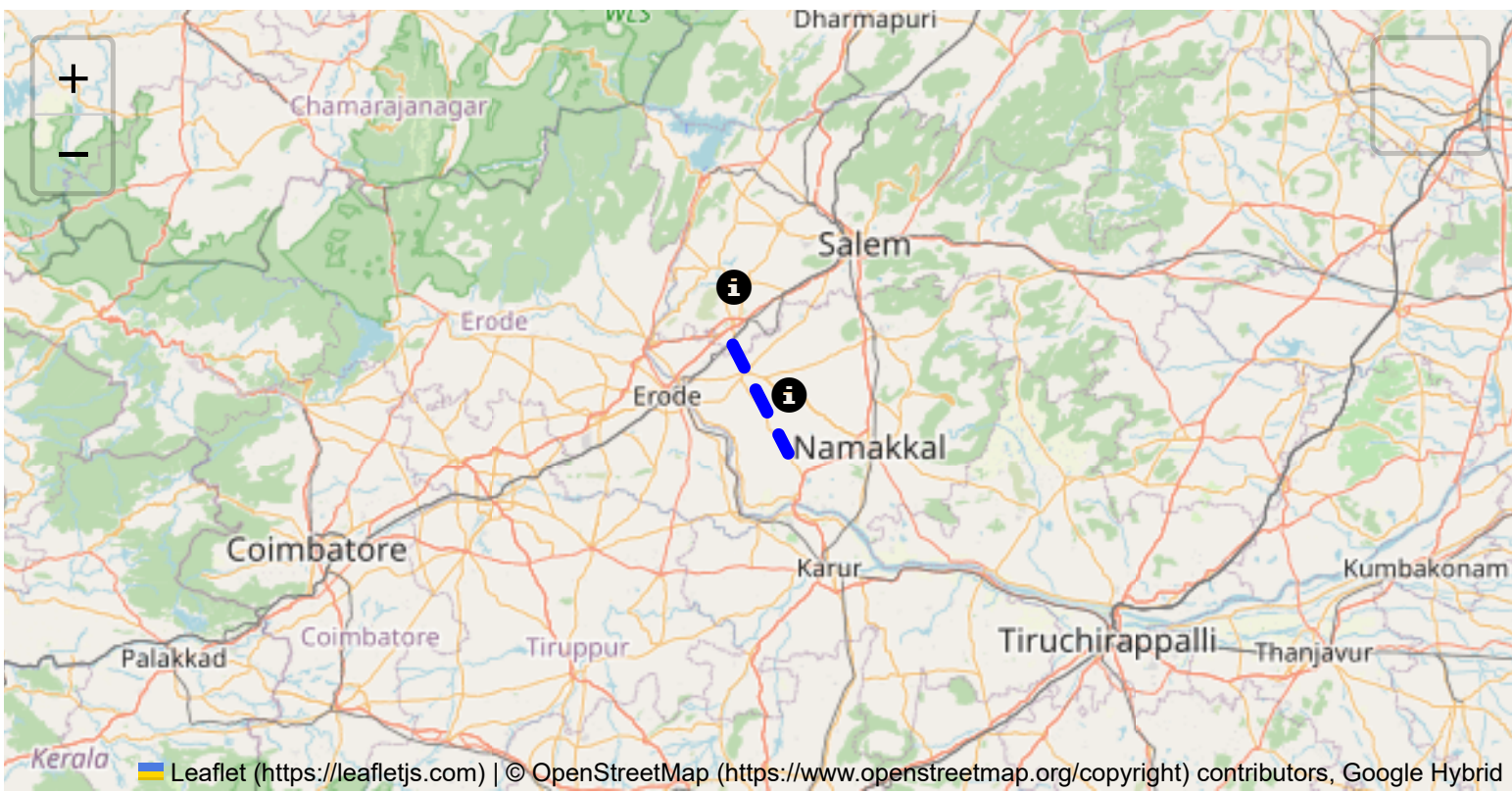
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

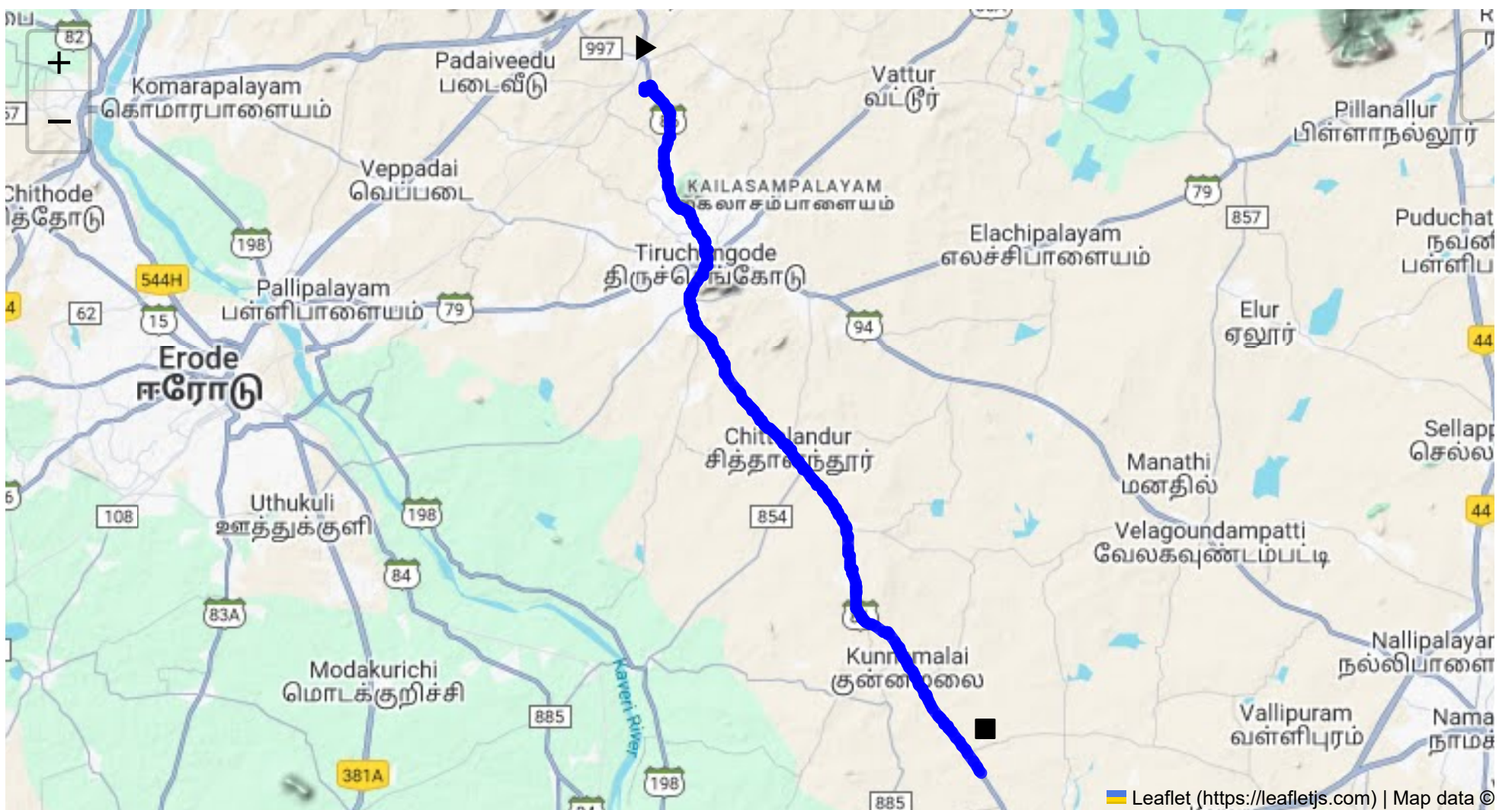
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 31.55 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.203955, 77.991753)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report: Sangagiri to Vasanthapuram, Tamil Nadu

1. Overview of the Route Map:

- The route spans approximately 31.55 kilometers from Sangagiri to Vasanthapuram. It typically involves navigating through both rural and semi-urban settings, with a mix of highway and local roads. Key roads may include stretches of NH544 or similar highways and rural connectors.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- **Weather:** The region typically experiences tropical climate with a dry season and a wet monsoon period (June to September).
- **Hazards:** Monsoon rains can lead to reduced visibility, slippery roads, and possible flooding in low-lying areas. During dry seasons, dust and glare from sun can also be factors.

3. Traffic Patterns:

- **Peak Hours:** Traffic is likely heavier during morning (7:30 AM - 9:30 AM) and evening (5:30 PM - 8:00 PM) due to commuter and local activity.
- **Congestion Areas:** Expect potential bottlenecks near markets or intersections on busy highways or entry/exit points to local towns.

4. Assessment of Road Quality and Infrastructure:

- **Road Quality:** Varies from well-maintained highways to potentially uneven or narrower rural roads.
- **Infrastructure:** Limited street lighting in some rural areas; watch for sharp bends, and absence of shoulders or barriers.

5. Suggestions for Alternative Routes for Emergencies:

- Identify nearby major highways or state roads (such as NH44) that can serve as detours. Ensure real-time navigation aids are active to provide updates.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Compliance with local transportation regulations is crucial, including permits for hazardous materials and adhering to vehicle and load safety standards. Night transportation of such materials might be restricted.

7. Overview of Historical Incidents:

- While specific incidents can vary, rural areas typically report fewer accidents than urban areas. Situations often involve rollovers, especially on curves or poorly maintained roads.

8. Environmental Considerations and Sensitive Areas:

- Watch for agricultural zones where open fields can contribute to visibility issues via windblown dust. Preserve nearby wildlife habitats by maintaining minimal noise and avoiding unnecessary off-road driving.

9. Analysis of Communication Coverage:

- Urban reaches and highways typically maintain adequate mobile coverage; however, certain rural stretches may experience dead zones, impacting navigation and emergency communication.

10. Estimated Emergency Response Times:

- **Urban/Highway Segments:** Response times could be within 20-30 minutes due to proximity of emergency services.
- **Rural Segments:** Expect longer response times, potentially 30-45 minutes or more, due to lower accessibility.

11. Overall Summary of Risk Assessment:

- This route presents moderate risk, with primary concerns being weather-related impacts during the monsoon, potential rural road quality issues, and communication gaps. By planning around peak traffic times and ensuring adherence to legal and safety frameworks for hazardous materials, these risks can be mitigated effectively. Regular updates on weather and traffic can significantly aid in maintaining safety.

In conclusion, careful pre-trip assessments and real-time adjustments will be key in maintaining a safe journey along this route for truck drivers carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453
